

The Mathematics of Geographic Arbitrage in Physical Commodity Markets

A Rigorous Treatment with a Synthetic-Data Laboratory

geo-arb project documentation

June 19, 2026

Abstract

Physical commodity merchants such as the large agricultural trading houses exist, in large part, to transform commodities across *space*, *time*, and *form*. This document gives a self-contained, rigorous account of the *spatial* (geographic) transformation: buying a commodity at one location and selling it at another. We formalise the delivered-cost identity, the spatial no-arbitrage band, and the Takayama–Judge spatial price equilibrium as a linear program whose dual variables are equilibrium location prices. We then specify a fully controlled stochastic data-generating process in which regional prices are *cointegrated* through a common world factor, freight and basis are mean-reverting, and FX follows a random walk. Within this process we derive the distribution of the arbitrage margin and the conditions under which a window opens. The companion **geoarb** Python package implements every object defined here, so each proposition can be checked numerically against simulation. **All data are synthetic; nothing here is investment advice.**

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1 Setup: the spatial market

We consider a single homogeneous commodity (say soybeans) traded across a finite set of locations partitioned into *origins* (export ports) and *destinations* (import ports).

Definition 1 (Spatial network). Let $\mathcal{O} = \{1, \dots, m\}$ index origins and $\mathcal{D} = \{1, \dots, n\}$ index destinations. A *lane* is an ordered pair $(i, j) \in \mathcal{O} \times \mathcal{D}$. Each lane has a great-circle distance $\delta_{ij} > 0$ (nautical miles) and a voyage time $\tau_{ij} = \delta_{ij}/(24v)$ days for a vessel of speed v knots.

At each trading date $t \in \{0, 1, \dots, T\}$ we observe:

- origin local cash prices $P_{i,t}^o$ and destination cash prices $P_{j,t}^d$ in USD per tonne;
- per-tonne freight $f_{ij,t}$ on each lane;
- port handling costs h_i^o, h_j^d ;
- an import tariff $\theta_{ij} \geq 0$;
- an FX rate $e_{i,t}$ (USD per unit of origin i 's local currency);
- a benchmark futures price H_t on a reference exchange.

2 Delivered cost and the arbitrage margin

The economic primitive is the *delivered cost*: the all-in per-tonne cost, expressed at the destination, of sourcing the commodity at origin i and landing it at destination j at the time the cargo arrives.

Definition 2 (Delivered cost). With financing rate r (annualised, 360-day convention), the delivered cost of lane (i, j) initiated at date t is

$$C_{ij,t} = \underbrace{\frac{P_{i,t}^o}{e_{i,t}} + h_i^o + f_{ij,t} + \theta_{ij} + h_j^d}_{\text{FOB cost}} + \underbrace{\left(\frac{P_{i,t}^o}{e_{i,t}} + h_i^o \right) r \frac{\tau_{ij}}{360}}_{\text{financing of cargo in transit}}. \quad (1)$$

Definition 3 (Arbitrage margin / window). The (gross) *arbitrage margin* of lane (i, j) at date t is

$$M_{ij,t} = P_{j,t}^d - C_{ij,t}. \quad (2)$$

The lane's *arbitrage window* is *open* at t if $M_{ij,t} > 0$ and *closed* otherwise.

The margin (2) is the per-tonne profit a trader books by buying FOB at i and selling at j today, ignoring price moves during transit; Section 7 restores that risk.

3 The spatial no-arbitrage band

Assumption 1 (Frictional law of one price). Capital is mobile and the number of potential shippers is large, so any lane with a strictly positive, riskless margin is exploited until the margin is competed away.

Proposition 1 (Spatial no-arbitrage band). *Under Assumption 1, in a frictionless competitive equilibrium with no binding capacity constraints,*

$$P_{j,t}^d - P_{i,t}^o/e_{i,t} \leq h_i^o + f_{ij,t} + \theta_{ij} + h_j^d + \left(\frac{P_{i,t}^o}{e_{i,t}} + h_i^o\right)r \frac{\tau_{ij}}{360} \quad \forall(i, j), \quad (3)$$

equivalently $M_{ij,t} \leq 0$ for every lane, with equality on every lane that carries strictly positive flow.

Proof. Suppose $M_{ij,t} > 0$ for some lane. A trader shipping one extra tonne earns $M_{ij,t} > 0$ at no risk, contradicting equilibrium under Assumption 1 unless the action is bounded. Competition drives $P_{j,t}^d$ down and $P_{i,t}^o$ up until $M_{ij,t} = 0$. Lanes that remain strictly unprofitable ($M_{ij,t} < 0$) carry no flow. Hence at equilibrium every active lane satisfies $M_{ij,t} = 0$ and every lane satisfies $M_{ij,t} \leq 0$, which is (3). $\square$ $\square$

Equation (3) is the *spatial* version of the Law of One Price: prices at two locations may differ, but only up to the full cost of moving product between them. The width of the band is precisely the right-hand side of (3) — the transfer cost — and *not* zero. This is why persistent price gaps between continents are not, by themselves, evidence of inefficiency.

4 Spatial price equilibrium as a linear program

When capacity binds and many origins compete for many destinations, the equilibrium flow pattern solves the classical Takayama–Judge transportation problem.

Definition 4 (Spatial price equilibrium, LP form). Given supplies s_i , demands d_j , and delivered costs C_{ij} at a fixed date, the competitive equilibrium flows $x_{ij} \geq 0$ solve

$$\begin{aligned} \max_{x \geq 0} \quad & \sum_{i \in \mathcal{O}} \sum_{j \in \mathcal{D}} (P_j^d - C_{ij}) x_{ij} \\ \text{s.t.} \quad & \sum_j x_{ij} \leq s_i \quad (\lambda_i) \\ & \sum_i x_{ij} \leq d_j \quad (\mu_j) \end{aligned} \quad (4)$$

where $\lambda_i, \mu_j \geq 0$ are the dual multipliers.

Theorem 1 (Duals are equilibrium prices). *Let $x^*, (\lambda^*, \mu^*)$ be primal–dual optimal for (4). Then complementary slackness gives, for every lane,*

$$P_j^d - C_{ij} \leq \lambda_i^* + \mu_j^*, \quad \text{with equality whenever } x_{ij}^* > 0. \quad (5)$$

The shadow price λ_i^ is the marginal value of an extra tonne of supply at origin i , and μ_j^* the marginal value of an extra tonne of demand at destination j . Active lanes earn exactly the spatial price difference implied by the duals; inactive lanes are weakly unprofitable.*

Proof. The Lagrangian of (4) is $\mathcal{L} = \sum_{ij} (P_j^d - C_{ij}) x_{ij} - \sum_i \lambda_i (\sum_j x_{ij} - s_i) - \sum_j \mu_j (\sum_i x_{ij} - d_j)$. Stationarity in $x_{ij} \geq 0$ (KKT) requires $P_j^d - C_{ij} - \lambda_i - \mu_j \leq 0$, with equality when $x_{ij} > 0$, which is (5). The envelope theorem gives $\partial(\text{optimal value})/\partial s_i = \lambda_i^*$ and $\partial/\partial d_j = \mu_j^*$, the stated marginal valuations. $\square$ $\square$

Theorem 1 makes Proposition 1 precise in the capacity-constrained case: the no-arbitrage band holds *lane by lane* with the duals replacing raw prices, and only the cheapest-to-serve origin–destination pairs carry flow. This is the mechanism by which competition “closes” arbitrage windows even when a naive scan of prices shows many positive margins.

5 A controlled stochastic data-generating process

To study *when* windows open we need a model of how prices, freight and FX co-move. We use a common-trend (factor) model that produces cointegrated regional prices, the empirically dominant feature of integrated commodity markets.

5.1 Prices: a common stochastic trend

Definition 5 (World factor and location prices). Let the world factor follow a random walk with drift,

$$F_t = F_{t-1} + \nu + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_F^2) \text{ iid.} \quad (6)$$

Each location price is the common trend plus a constant intercept a_k plus a stationary idiosyncratic spread $u_{k,t}$:

$$P_{k,t} = F_t + a_k + u_{k,t}, \quad u_{k,t} = \rho u_{k,t-1} + \varepsilon_{k,t}, \quad |\rho| < 1, \quad (7)$$

with $\varepsilon_{k,t} \sim \mathcal{N}(0, \sigma_u^2)$ iid across k, t .

Proposition 2 (Regional prices are cointegrated). *For any two locations k, ℓ , the spread $Z_t := P_{k,t} - P_{\ell,t}$ is (covariance-)stationary; equivalently P_k and P_ℓ are cointegrated $CI(1, 1)$ with cointegrating vector $(1, -1)$.*

Proof. From (7), $Z_t = (a_k - a_\ell) + (u_{k,t} - u_{\ell,t})$. The common trend F_t cancels. The difference of two independent stationary AR(1) processes is stationary, so Z_t is stationary with mean $a_k - a_\ell$. Each $P_{k,t}$ is individually $I(1)$ because F_t is a random walk. A stationary linear combination of $I(1)$ series is cointegration by definition. $\square$ $\square$

Corollary 1 (Mean reversion and half-life). *The demeaned spread $\tilde{Z}_t := Z_t - (a_k - a_\ell)$ obeys an AR(1) with coefficient ρ , so it mean-reverts with half-life*

$$t_{1/2} = \frac{\ln(1/2)}{\ln |\rho|}. \quad (8)$$

The package function `metrics.cointegration_check` estimates ρ by OLS and reports $t_{1/2}$; on the default calibration ($\rho = 0.92$) the predicted half-life is $\ln(0.5)/\ln(0.92) \approx 8.3$ days, matching the simulation output (8.2 days).

5.2 Freight and basis

Freight is volatile but does not trend without bound; we model log-freight as a seasonal mean-reverting (Ornstein–Uhlenbeck/AR(1)) process:

$$\ln f_{ij,t} = \ln \bar{f}_{ij} + A \sin\left(\frac{2\pi t}{252}\right) + g_{ij,t}, \quad g_{ij,t} = \phi g_{ij,t-1} + \xi_{ij,t}. \quad (9)$$

The local *basis* relative to the benchmark future is $b_{k,t} := P_{k,t} - H_t$, where $H_t = \bar{H} + F_t + ct$ embeds a deterministic carry c . Because H_t shares the world factor F_t , the basis $b_{k,t} = a_k + u_{k,t} - \bar{H} - ct$ inherits stationarity around its (carry-adjusted) mean.

6 Distribution of the arbitrage margin

We can now describe the margin's law. Hold FX at $e_{i,t} \approx 1$ and write the transfer cost as $K_{ij,t} = h_i^o + f_{ij,t} + \theta_{ij} + h_j^d + (\text{financing})$.

Proposition 3 (Margin decomposition). *The arbitrage margin admits the decomposition*

$$M_{ij,t} = \underbrace{(a_j - a_i) - \bar{K}_{ij}}_{\text{structural mean } \bar{M}_{ij}} + \underbrace{(u_{j,t} - u_{i,t})}_{\text{stationary spread}} - \underbrace{(f_{ij,t} - \bar{f}_{ij})}_{\text{freight shock}} + (\text{financing, tariff, FX terms}), \quad (10)$$

where $\bar{K}_{ij} = \mathbb{E}[K_{ij,t}]$. Consequently $M_{ij,t}$ is stationary; the common world trend F_t does not enter, so margins do not inherit the nonstationarity of price levels.

Proof. Substitute (7) and Definition 2 into (2); the F_t terms in $P_{j,t}^d$ and $P_{i,t}^o$ cancel exactly as in Proposition 2. The remaining components are stationary (AR(1) spreads, mean-reverting freight) or deterministic, so $M_{ij,t}$ is stationary. $\square$ $\square$

Corollary 2 (Window-open probability). *If the stationary part of $M_{ij,t}$ is approximately Gaussian with mean $\bar{M}_{ij}$ and standard deviation $\sigma_{M_{ij}}$, the long-run fraction of days the window is open is*

$$\Pr(M_{ij,t} > 0) \approx \Phi\left(\frac{\bar{M}_{ij}}{\sigma_{M_{ij}}}\right), \quad (11)$$

where Φ is the standard normal CDF. Structurally short, cheap lanes ($\bar{M}_{ij} > 0$) are open most of the time; structurally unprofitable lanes ($\bar{M}_{ij} < 0$) open only on favourable spread/freight shocks.

This corollary is the quantitative content of “the arb is mostly closed”: for a lane with $\bar{M}_{ij} < 0$, windows are tail events driven by transitory dislocations in the basis spread or freight.

7 Trading under transit risk and the futures hedge

A cargo booked at t is sold only on arrival at $t + \tau_{ij}$. The *realised* per-tonne PnL is therefore

$$\Pi_{ij,t} = P_{j,t+\tau_{ij}}^d - C_{ij,t}, \quad (12)$$

which differs from the signal $M_{ij,t} = P_{j,t}^d - C_{ij,t}$ by the destination price move during transit, $P_{j,t+\tau}^d - P_{j,t}^d$. Since P_j^d contains the random walk F , this move has variance growing in τ — the *flat-price* risk of a physical position.

Proposition 4 (Futures hedge isolates the spatial spread). *Sell one tonne of the benchmark future at H_t when booking the cargo and buy it back at arrival. The hedged PnL is*

$$\Pi_{ij,t}^h = (P_{j,t+\tau}^d - C_{ij,t}) + (H_t - H_{t+\tau}) = \underbrace{(b_{j,t+\tau}^d - (C_{ij,t} - H_t))}_{\text{basis / spatial spread}} - c\tau, \quad (13)$$

so the world-factor term F cancels between P_j^d and H , leaving a stationary basis spread plus a deterministic carry. The variance of Π^h does not grow with τ , unlike the unhedged Π .

Proof. Write $P_{j,t+\tau}^d = H_{t+\tau} + b_{j,t+\tau}^d$ by definition of basis. Substituting into Π^h cancels $H_{t+\tau}$: $\Pi^h = H_{t+\tau} + b_{j,t+\tau}^d - C_{ij,t} + H_t - H_{t+\tau} = b_{j,t+\tau}^d - (C_{ij,t} - H_t)$. With $H_t = \bar{H} + F_t + ct$ the residual carry is $-c\tau$; all F terms have cancelled. Basis is stationary (Section 5), so $\text{Var}(\Pi^h)$ is bounded in τ . $\square$ $\square$

Proposition 4 is the formal statement of why physical arbitrage desks hedge flat price with futures and trade the *basis*: the hedge removes the unbounded random-walk risk and leaves exactly the mean-reverting spatial spread the trader has an informational edge on. The simulator (`simulator.ArbitrageSimulator`) implements both the unhedged and hedged PnL so the variance reduction can be measured directly.

8 From theory to the geoarb package

Object in this paper	Equation	Code
Network, voyage time	Def. 1	<code>geography.py</code>
Delivered cost $C_{ij,t}$	(1)	<code>arbitrage.delivered_cost</code>
Arbitrage margin $M_{ij,t}$	(2)	<code>arbitrage.ArbitrageEngine</code>
No-arbitrage band	Prop. 1	<code>ArbitrageEngine.window_stats</code>
Spatial LP & duals	(4), Thm. 1	<code>equilibrium.SpatialEquilibrium</code>
Common-trend prices	(7)	<code>dgp.SyntheticDGP</code>
Cointegration / half-life	Prop. 3, Cor. 1	<code>metrics.cointegration_check</code>
Margin distribution	Prop. 4, Cor. 2	<code>ArbitrageEngine.window_stats</code>
Hedged PnL	Prop. 5	<code>simulator.ArbitrageSimulator</code>

Table 1: Correspondence between the formal objects and the implementation.

9 Caveats and what the synthetic laboratory does *not* prove

The simulator routinely reports high Sharpe ratios. This is a feature of the *idealised* environment, not a claim about real markets. The synthetic laboratory deliberately omits: execution slippage and market impact; counterparty and demurrage risk; discrete, lumpy vessel availability; information asymmetry and adverse selection; credit and margin calls on the futures leg; and the fact that real basis is only partially predictable. Each omission, once restored, compresses the Sharpe toward the thin, hard-won margins that characterise the real physical-trading business. The value of the laboratory is pedagogical and structural: it lets one see *exactly* which mechanism (cointegration, the no-arb band, the LP duals, the futures hedge) produces each observed feature, because every mechanism is written down in closed form.

References (selected, for further reading)

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Disclaimer. All prices, freight rates, FX, and flows in the companion software are synthetic and generated by the process of Section 5. This document is educational and is not investment, financial, or trading advice.